Load Identification Based on Attention Semisupervised Curriculum Label Learning With AVME-HT Feature

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***Abstract*— Nonintrusive load monitoring (NILM) is a cost-effective technology for monitoring detailed electricity energy consumption. In recent years, machine learning has emerged as the predominant approach for achieving NILM tasks. However, machine learning models require an extensive volume of labeled data which is unachievable in the real world. Additionally, the collected power data is susceptible to noise interference arising from grid or measurement error. To this end, this article presents a new NILM method, adaptive variational mode extraction-Hilbert transform-excitation attention-semisupervised curriculum label learning (AVME-HT-EA-SCLL), which offers robust antinoise performance, comprehensive feature extraction, efficient utilization of unlabeled data, and exceptional classification accuracy. The main contributions of this work include the proposed adaptive variational mode extraction (AVME) method for efficient signal decomposition, a new feature extraction method, which combines AVME and Hilbert transform to extract unique features and filter out noise interference, a semisupervised learning (SSL) model based on curriculum learning strategy to identify appliances with limited labeled data, and an excitation attention (EA) network to amplify attention toward crucial information. Experimental evaluations conducted on public datasets such as the plug-load appliance identification dataset (PLAID) and the worldwide household and industry transient energy dataset (WHITED), as well as a private dataset collected in the laboratory, demonstrate that the proposed method surpasses state-of-the-art approaches.**

***Index Terms*— Curriculum learning strategy, excitation attention (EA), load identification, nonintrusive load monitoring (NILM), semisupervised learning (SSL), variational mode extraction.**

I. Introduction

ATTAINING carbon peak and carbon neutrality requires a comprehensive and significant transformation of economic and social systems. The primary concern of the dual carbon objectives is enhancing energy efficiency. Conventional power systems are undergoing a significant transformation, transitioning from fossil fuel generation to a renewable energy-centric structure. Digitization is undoubtedly a critical facilitator of the clean energy transition, driving the change and laying the groundwork for efficient energy monitoring.

Nonintrusive load monitoring (NILM) is a technology that enables the identification and monitoring of electrical appliances and their energy consumption patterns using nonintrusive sensors and data analysis techniques. Hart [1] first proposed the concept of NILM, which aims to identify the operational state (on/off) and the precise power consumption of individual electrical loads, considering as input only the aggregated consumption of these loads [2]. The complexity and cost of the NILM solution are independent of the number of appliances to be monitored. Hence, the technology has massive potential for widespread deployment in smart residential buildings [3].

For load identification, distinctive features are derived from recorded load signatures, including current, voltage, power, or their combinations. The performance of load identification hinges on two key aspects, namely, the feature extraction process and the classifier design. The feature extraction process involved methods considering image features such as voltage–current (V–I) trajectory [4] or time series features [5], [6] from load signatures. To improve V–I trajectory images, weighted pixelated images of V–I trajectories were proposed [7], which utilizes thresholds determined through particle swarm optimization (PSO). Although this method improves accuracy, the application of the PSO increases the computational complexity of feature extraction. Furthermore, short-time Fourier transform (STFT) [8] is performed rather than directly considering V–I trajectories. Specifically, the method computed the root mean square (rms) of transient currents at 1-s intervals. These rms values are transformed into time–frequency domain images using STFT, with an input size of 101×26. However, the precise capturing requirement for transient currents in this method increases the difficulty and complexity of the feature extraction process. Additionally, owing to the limited information present in transient currents, the resulting features fail to comprehensively capture load characteristics.

In terms of the time series feature, empirical mode decomposition (EMD) in [9] was employed in load disaggregation to decompose the signal of each electric appliance into a series of intrinsic modal function (IMF) components. However, EMD

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might overly decompose signals, resulting in an excessive or ambiguous number of generated IMFs. Additional algorithms are needed to determine the effectiveness of generating IMF, increase the complexity of the feature extraction process, and reduce the performance of load decomposition. Mitra et al. [10] employed the variational mode decomposition (VMD) method, auto-correlogram analysis, and wavelet packet tree decomposition to extract a comprehensive array of features from current signals. The input data comprised 1500 load current samples. A set of features with a dimensionality of 120 was derived. However, the high dimensionality of the features amplifies the computational complexity of the feature extraction process. Existing methods, whether using image-based or time series features, have shown promising results but add complexity to the feature extraction process. Moreover, they overlook the impact of noise interference. In real-world scenarios, power signals are susceptible to external environmental noise, which can amplify noise effects in the resulting features, obscuring load characteristics and compromising classification performance.

The application of the deep learning model as the utilized classifier in NILM has gained significant attention in recent years. These classifiers are mainly grouped into three categories: supervised, unsupervised, and semisupervised models. Supervised models require labeled data for each individual appliance. The first NILM problem was solved using the supervised learning model by Kelly and Knottenbelt [11]. Subsequently, models based on convolutional neural network (CNN) and long short-term memory (LSTM) recurrent neural networks were proposed for load classification [12], [13], [14]. The classifiers based on supervised learning models demonstrate high accuracy, yet the availability of annotated label samples is limited, posing a challenge to their practical application. On the other hand, unsupervised models learn to distinguish between appliances, devoid of prior information or employing manual labeling. Liu et al. [15] designed a fuzzy clustering algorithm called entropy index constraints competitive agglomeration for low-complexity unsupervised NILM. An unsupervised NILM method [16] was proposed for identifying electric bicycle charging loads and assessing the status of the underlying batteries. Although unsupervised learning does not require actual labels, it is incapable of determining the types of appliances. Additional information is necessary to identify load categories. Moreover, the performance of unsupervised learning algorithms tends to decrease with an increase in the number of appliances.

As a trade-off between supervised and unsupervised learning, semisupervised learning (SSL) can learn from limited labeled data and a large number of unlabeled samples, which is more suitable for practical NILM applications. An SSL model based on student–teacher framework [17] was employed for load identification. Nonetheless, the efficacy of this framework highly relies on the precision of the teacher model. The teacher model's inaccuracies due to inadequate training on limited labeled data can propagate errors within this framework. A graph-based multilabel SSL method [18] was proposed to perform load identification. Additionally, a Co-training method using two classifiers was presented in [19] for SSL-based NILM. However, this method requires both classifiers to have identical predicted results to label unlabeled data, imposing a stringent criterion that diminishes the utility of unlabeled data. Although these SSL methods can utilize unlabeled observations, they fundamentally rely on a fixed criterion to label unlabeled data. To improve load identification performance, additional SSL strategies are required to optimize the utilization of unlabeled data for enhanced load identification.

To address the limitations of the existing literature mentioned above, a novel semisupervised model called adaptive variational mode extraction-Hilbert transform-excitation attention-semisupervised curriculum label learning (AVME-HT-EA-SCLL) was proposed for load identification. Specifically, to mitigate the complexity of the feature extraction process, an efficient signal decomposition method, adaptive variational mode extraction (AVME), was proposed. Furthermore, to reduce the impact of noise interference during the feature extraction process and enhance the classification accuracy, an AVME-HT feature was introduced. Finally, an SSL model based on a curriculum learning strategy and the excitation attention (EA) network was proposed to improve the classification performance with limited labeled data. The key contributions are summarized as follows.

1) As current signal decomposition techniques such as VMD are suffering from high computational complexity, an efficient algorithm, AVME, was proposed in the context of NILM. The proposed AVME method employed the Pauta criterion to intelligently tune the algorithm's parameters, resulting in low computational complexity and significant signal decomposition performance.
2) A new feature extraction method was proposed to perform both signal denoising and feature extraction simultaneously. The energy loss coefficient, Hilbert transform, and AVME are explored to generate AVME-HT features with strong distinguishability and noise resistance, enhancing the performance of load classification.
3) To improve the load classification performance with limited labeled samples, an effective semisupervised model based on the curriculum learning strategy was presented. The model dynamically adjusts pseudolabel thresholds based on varying learning difficulties across different categories, optimizing the utilization of unlabeled data.
4) The EA network was introduced to assign weights to different channels of the AVME-HT features, aiding the model in discerning critical information. The incorporation of the semisupervised model with the EA network can better extract crucial feature information from limited labeled data and improve understanding of unlabeled data's structure, enhancing classification performance.

II. Relate Work

This section delves into the pertinent literature concerning the methodology presented in this study, encompassing signal decomposition, SSL, and attention mechanism. In recent years, various signal processing methodologies such as EMD and VMD have been introduced in the field of NILM. Saleh et al. [20] proposed an EMD-based filtering algorithm for signal

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analysis that can extract useful information about individual appliance consumption. However, EMD is susceptible to mode mixing issues, making it challenging to accurately separate desired modes. Ghosh et al. [21] introduced a novel feature extraction method for NILM. This method involves processing 0.1 s of current data using VMD and autocorrelation. It computes three modal signals and their corresponding auto-correlogram signals, each consisting of 6000 samples. Subsequently, statistical features of the signals are calculated, resulting in a total of 60-D features. Finally, feature selection is performed through three feature ranking criteria. However, the entire feature extraction process is relatively complex. Cui et al. [22] introduced an improved VMD method designed for denoising training data associated with industrial park loads. The input data comprised load data from textile enterprises, collecting 96 data points at 15-min intervals. The improved VMD was utilized to extract four modes from the load data at 20, 30, and 40 signal-to-noise ratio (SNR). However, this method's computational complexity is notably elevated, which, in turn, diminishes the efficiency of the denoising process. Existing VMD-based methods still face challenges related to high computational complexity and extended computation times, significantly affecting feature extraction and data denoising.

SSL has tremendous value in practice due to the utilization of both labeled and unlabeled data. Yang et al. [23] introduces a framework based on semisupervised multilabel deep learning to decrease reliance on extensive labeled datasets, but its effectiveness varies with the labeling rate. A new multilayer momentum contrast learning mechanism is introduced for the purpose of self-supervised feature representation learning [24]. A graph-based SSL [25] was employed to annotate unlabeled historical events as part of the process for recognizing multienergy coupling appliances. It is worth noting that these SSL algorithms often rely on a fixed threshold to filter out unlabeled data and treat all classes equally, regardless of their different learning difficulties. Consequently, further research is needed to develop more advanced SSL methods that can adaptively handle different classes and utilize unlabeled data more effectively for improved load recognition in NILM.

Attention mechanisms improve classification performance by allowing the network to focus on specific parts of the input data that are more important for the current task. Yin and Ma [26] introduced an attention-based CNN to learn V–I images for load identification, but it involves weight calculations for each image, increasing computational complexity. A multiscale self-attention network was proposed [27] to utilize the global temporal correlation and local sequential features for energy disaggregation tasks. The introduction of a novel model named InFocus incorporated a self-attention mechanism into convolutional layers, enabling the model to allocate attention weights to various features [28]. Self-attention mechanism demands substantial data for effective pattern learning, rendering them less efficient with limited training data. Therefore, applying attention mechanisms to NILM issues still faces significant challenges.

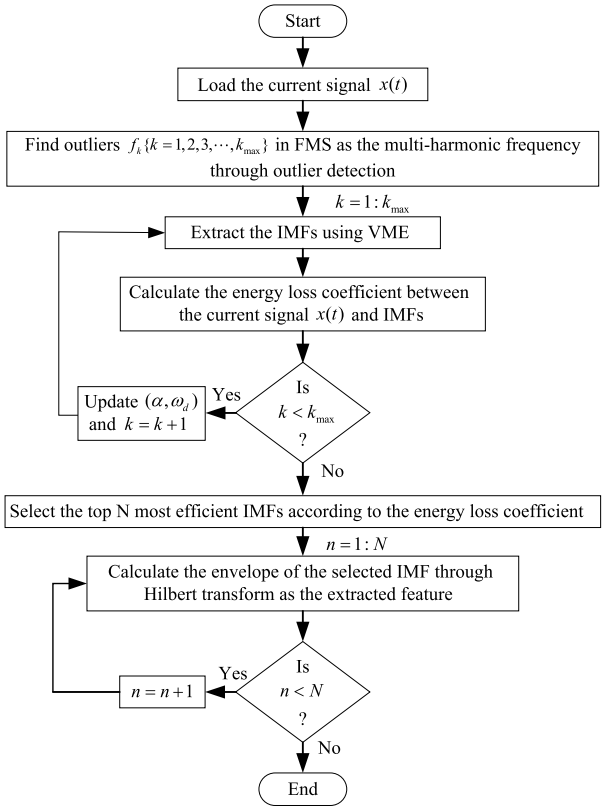


Fig. 1. Flowchart of the proposed feature extraction method.

III. Feature Extraction Based on Adaptive Variational Mode Extraction

The feature extraction process in the existing methods is relatively complex and susceptible to noise interference. To this end, we introduce AVME, a new feature extraction technique that efficiently decomposes the signal into several desired modes as 1-D features. The proposed method can reduce the complexity of feature extraction process, enhance feature significance, and perform signal denoising. Section III-A introduces the existing signal decomposition algorithms, VMD and VME, while emphasizing their limitations. Section III-B outlines the improvements made by the proposed AVME method over VME and elucidates how it can be applied to feature extraction in NILM. The flowchart depicting the proposed feature extraction method is presented in Fig. 1.

A. Signal Decomposition Methods: VMD and VME

The VMD method [29] is currently one of the most popular signal decomposition methods. It incorporates various signal processing techniques, including Wiener filtering for denoising, Hilbert transform to create single-sideband analytic signals, and complex harmonic mixing for frequency shifting to the baseband within its mathematical framework. The candidate mode is first single-sided, then shifted to baseband, and finally L^2-norm of the H^1 Gaussian smoothness is calculated

$$\mathrm{BW}_k = \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2 \quad (1)$$

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where ∂ symbolizes the Dirac distribution, and $*$ signifies convolution. Furthermore, u_k and ω_k denote the kth mode and its center frequency, while BW_k can be regarded as an indicator of the mode's bandwidth. To achieve maximal separation modes while decomposing the input signal $f(t)$ into modes $u_k(t)$, the VMD approach addresses the following optimization problem:

$$\min_{\{u_k\},\{\omega_k\}} \left\{ \alpha \sum_k \mathrm{BW}_k^2 + \left\| f(t) - \sum_k u_k(t) \right\|_2^2 \right\} \tag{2}$$

where the first term is employed to enforce compactness among the modes and the second term is the reconstruction error which should be minimized. α is a weighting parameter to balance the compactness of the modes and the reconstruction error. $f(t)$ is the input signal. As long as the number of modes k is set, VMD can automatically decompose the signal into the corresponding number of modes. However, the iterative optimization process employed by VMD to determine the central frequency of each decomposed mode continuously adjusts and updates until achieving optimal separation of k modes. This procedure augments the computational complexity of feature extraction.

Nazari introduced the VME method, which is established on the same concepts utilized in VMD: Wiener filtering, the HT, and a variational approach. VME extracts a desired mode by offering an initial estimate of the mode's center frequency. VME does not require iterative optimization process to determine the central frequency of modes, as required by VMD. Therefore, it significantly reduces the computational complexity in extracting a specific mode. From [30], it is evident that the final derivation of the VME method leads to the iterative update of the following equation:

$$\hat{u}_d^{n+1}(\omega) = \frac{\hat{f}(\omega) + \alpha^2\left(\omega - \omega_d^n\right)^4 \hat{u}_d^n(\omega) + \frac{\hat{\lambda}(\omega)}{2}}{\left[1 + \alpha^2\left(\omega - \omega_d^n\right)^4\right]\left[1 + 2\alpha\left(\omega - \omega_d^n\right)^2\right]} \tag{3}$$

$$\omega_d^{n+1} = \frac{\int_0^\infty \omega \left|\hat{u}_d^{n+1}(\omega)\right|^2 d\omega}{\int_0^\infty \left|\hat{u}_d^{n+1}(\omega)\right|^2 d\omega} \tag{4}$$

$$\hat{\lambda}^{n+1} = \hat{\lambda}^n + \tau \cdot \left[\frac{\hat{x}(\omega) - \hat{u}_d^{n+1}(\omega)}{1 + \alpha^2\left(\omega - \omega_d^{n+1}\right)^4} \right] \tag{5}$$

where $\hat{u}_d^n(\omega)$ is the obtained desired mode in the nth iteration with the center frequency ω_d^n, $\hat{x}(\omega)$ represents the Fourier transform of the original signal, and α is the weighting coefficient that controls the mode compactness. λ refers to the Lagrangian multiplier. n denotes the number of iterations, and τ is the iteration step length. It can be observed from (3) and (4) that the estimated value of the center frequency ω_d is a necessary input parameter for VME to extract the desired mode $\hat{u}_d(\omega)$. Furthermore, it is also unable to distinguish whether the modes extracted by the VME method are effective modes or noise modes. Therefore, this article proposed the AVME method to address the limitations of VME.

B. Proposed Adaptive Variational Mode Extraction

The proposed AVME addresses the limitations of the existing signal decomposition methods and efficiently applies it to feature extraction. Fig. 2(a)–(c) illustrates the mode decomposition of the current signals of three loads in the presence of 10-dB SNR noise, performed using the proposed AVME method. The *Signal* shown represents the load current contaminated with 10 dB noise, while the IMF refers to the decomposed modes obtained through the AVME algorithm. This decomposition process helps to separate the load current into different harmonic modes, enabling a more detailed analysis and characterization of the electrical behavior of each load. The specific details of this method are as follows.

1) Determination of Mode Center Frequencies: The center frequency ω_d is a crucial input parameter in the VME method. However, the current signals of various types of loads contain different harmonic modes, which makes it challenging to determine the center frequency of each mode. To overcome this limitation, the proposed AVME method utilizes an outlier detection method, the Pauta criterion, for the rapid localization of center frequencies for each mode, eliminating the need for manual parameter setting.

Initially, the spectrum of the signal is obtained via Fourier transform, revealing that the center frequency of different modes is the local maximum of outliers $f(k)\{k = 0, 1, 2, \ldots, k_{\max}\}$, as illustrated in Fig. 3. To locate these outliers, an outlier detection method, the Pauta criterion, combined with the sliding window is employed due to its lower computational complexity. The process involves iterative computation of the following equation:

$$|f(k_{\text{outlier}})| > \mu(|f(k_i)|) + 3 \cdot \sigma(|f(k_i)|) \tag{6}$$

where $\mu(\cdot)$ and $\sigma(\cdot)$ indicated the average value and standard deviation, respectively. $f(k_i)$ represented the frequency in the ith sliding window with length W_1. $f(k_{\text{outlier}})$ is the frequency corresponding to the outlier ($k_{\text{outlier}} \in \{k_i | W_1 \cdot (i-1) + 1 \leq k \leq W_1 \cdot i\}$). The detected outliers are stored and replaced with the arithmetic mean of the adjacent frequencies, and the process is repeated until no more outliers are found. Finally, the central frequency of the modes is determined by computing the local maxima of these outliers. Therefore, as depicted in (6), the proposed AVME method utilizes outlier detection to effectively ascertain the center frequencies of various modes in the signal. This approach addresses the limitations of VME, significantly reducing the computational complexity.

2) Identification of Effective Modes: The AVME method extracts IMFs comprising signal-dominant and noise-dominant modes. To effectively extract features and denoise the signal, the proposed method utilizes the energy loss coefficient to identify the most informative IMFs. Initially introduced in [31], the energy loss coefficient is defined as the ratio of the energy of the decomposition residual to the energy of the original signal. It can be calculated using the following formula:

$$e = \frac{\left\| F_{\text{original}} - F_k \right\|_2^2}{\left\| F_{\text{original}} \right\|_2^2} \tag{7}$$

where F_k denotes the extracted IMF and F_{original} represents the original signal. As indicated in (7), lower energy loss

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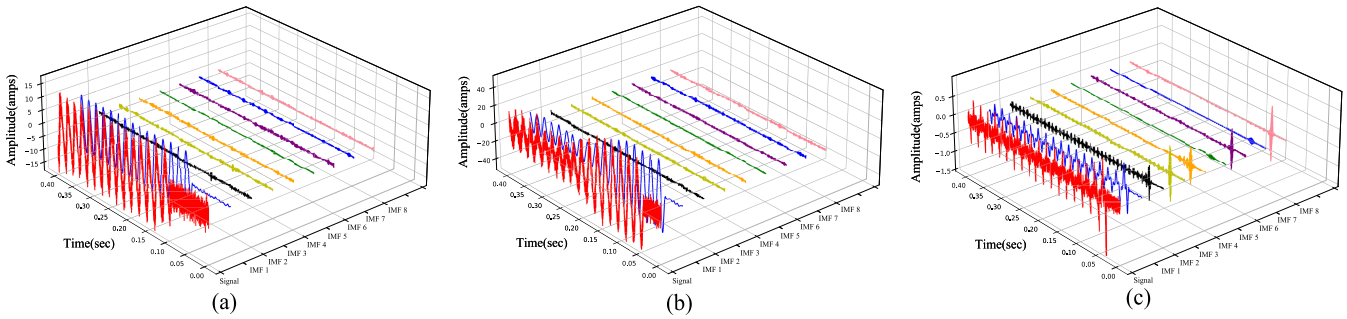


Fig. 2. (a)–(c) Mode decomposition of different loads using the proposed AVME method.

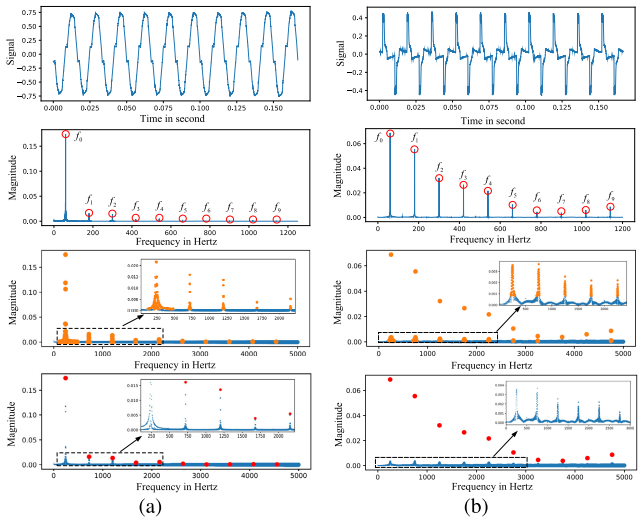


Fig. 3. (a) and (b) Schematic of the determining center frequencies through outlier detection.

coefficients imply higher correlation and more valuable information. Therefore, selecting IMFs with low energy coefficients to determine valuable modes addresses the limitations of the VME method in terms of data denoising. Through multiple experiments, the first five IMFs with the lowest coefficients were chosen for their uniqueness and anti-interference ability in feature extraction.

3) Hilbert Transform: The envelopes of the IMFs were calculated using the HT to enhance the discriminatory power of the features. The calculation can be performed using the following expression:

$$e(F_k) = |F_k + \text{Hilbert}(F_k)| \tag{8}$$

where $|\cdot|$ represents the modulo operation and $e(F_k)$ indicate the AVME-HT features. In this study, the AVME-HT feature consists of 5 channels, each with a length of 1000 samples.

IV. SemiSupervised Curriculum Label Learning

The aim of SSL is to improve model generalization by utilizing unlabeled data. However, current SSL methods frequently depend on a single criterion, lacking adaptable strategies for leveraging unlabeled data. Addressing this limitation, this study presents an attention-based semisupervised model that combines curriculum learning with an EA network.

Algorithm 1 Curriculum Learning Strategy

Input: labeled data X, unlabeled data U.

$T_s = 100\% : s \in (1, \ldots, S)$ {Initialize the thresholds.}
$\Delta := 10\%$
$\Psi = 100\% - \Delta$
while not reach the maximum iteration, or U is empty **do**
  $P_\theta^t \leftarrow$ train classifier using X
  $G \leftarrow$ infer classifier P_θ^t using U
  for $n = 1; n < N; n++$ **do**
    if $P_\theta^t(u_n) > T_s$ **then**
      $X = X \cup (u_n, pseudo - label(p_\theta^t, u_n))$
    end if
  end for
  for $s = 1; s < S; s++$ **do**
    $T_s = Percentile(G_s, \Psi)$
  end for
  $\Psi = \Psi - \Delta$
end while

Return: Model parameters.

The proposed model dynamically adapts learning strategy to optimize the utilization of unlabeled data, leading to enhanced load identification.

A. Pseudolabeling Under Curriculum Learning Strategy

1) Pseudolabeling: Our approach is based on the self-training framework, where a classifier undergoes multiple iterations of training. In each iteration, the classifier leverages predictions from the previous iteration to relabel unlabeled samples. The predictions made on unlabeled samples are also referred to as pseudolabels. These pseudolabels are then used to augment the labeled subset and train a new classifier. The process is repeated until the maximum iteration is reached or all unlabeled subsets are labeled with pseudolabels. In our approach, we introduce a curriculum learning strategy to gradually select a subset of unlabeled samples along with their corresponding pseudolabels.

2) Curriculum Learning Strategy: The fundamental concept of curriculum learning involves beginning with simpler examples and progressively advancing to more challenging ones. This approach is motivated by the observation that humans tend to learn more effectively when examples are presented in an ordered sequence of difficulty rather than

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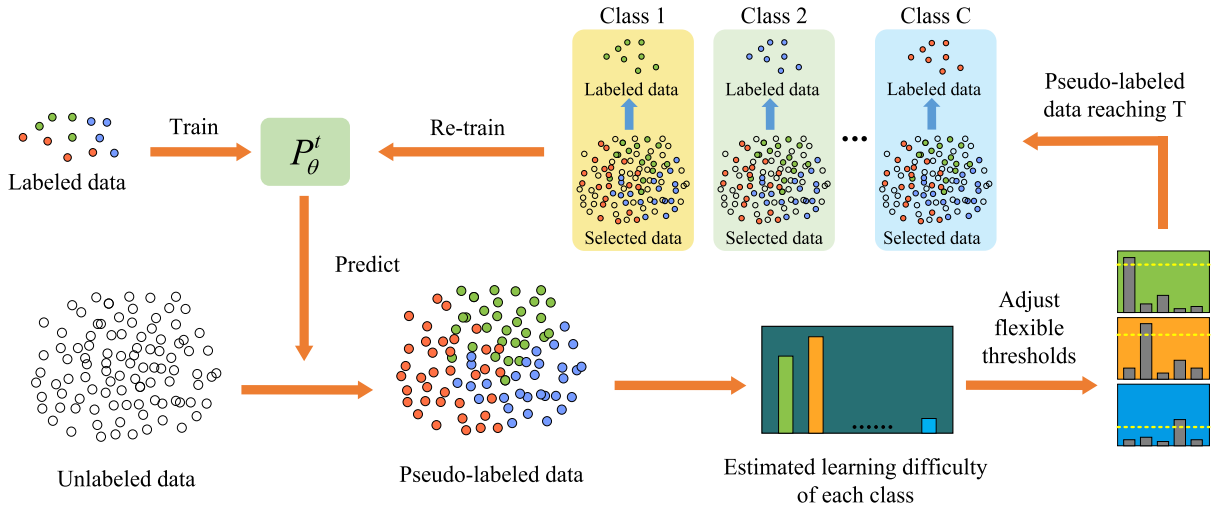


Fig. 4. Illustration of curriculum learning strategy. Initially, the labeled dataset is utilized to train the classifier P_θ^t. Subsequently, this classifier predicts and relabels unlabeled samples. The probability distribution of the highest predictions on unlabeled samples is then employed to gauge the learning difficulty of each class and fine-tune the thresholds accordingly. The pseudolabel samples are used to augment the labeled subset and train a new classifier. This process is reiterated until all unlabeled data are pseudolabeled or the maximum iteration limit is attained.

randomly [32], [33]. In NILM, the SSL model should not treat all classes equally but rather take into account the diverse learning challenges arising from the distinctive characteristics of each load. The model should be trained in a sequential manner, progressing from simpler to more complex tasks, guided by the varying levels of difficulty associated with each load.

To do so, a curriculum learning strategy was proposed. Algorithm 1 presents the pseudocode for the proposed strategy. The pseudocode defines the labeled data $X = \{(x_m, y_m) : m \in (1, \ldots, M)\}$ and unlabeled data $U = \{u_n : n \in (1, \ldots, N)\}$, with x, y, M, u, and N representing labeled data, corresponding labels, labeled data size, unlabeled data, and unlabeled data size, respectively. T_s denote the flexible threshold for class s. The initial value of T_s is set to 100%. P_θ^t is the classifier with parameter θ. $G = [G_1, \ldots, G_s, \ldots, G_S]$ represents the pseudolabels for different classes on unlabeled data U.

First, the model P_θ^t is trained only on the labeled data X during the first iteration. It then performs inference on the unlabeled data U, obtaining the pseudolabels G for each unlabeled sample. Second, samples with pseudolabels exceeding the threshold T_s are incorporated into X, while the remaining samples are returned to U. Third, T_s are updated according to the pseudolabels G_s from this iteration. The following percentile function was employed:

$$T_s = \text{Percentile}(G_s, \Psi) = G_s[(\Psi/100) * (K + 1)] \quad (9)$$

where $\text{Percentile}(G_s, \Psi)$ is defined as the Ψth percentile of G_s. K denotes the total number of G_s. It can be observed from (9) that the Ψth percentile of G_s is used as the updated T_s. In other words, the thresholds are adjusted according to the probability distribution for the highest predictions on unlabeled samples. With each iteration, the percentile decreases, starting from the 90th percentile in the initial iteration to the 80th percentile, then the 70th percentile, and so on. This process continues by reducing the previously chosen percentile by 10% with each iteration. Consequently, all unlabeled data were incorporated after ten iterations.

With the increase in iterations, the updated threshold gradually decreases. This is because the model initially needs to learn from high-confidence samples, which represent simple tasks, enabling the model to learn correctly. With further iterations, more challenging tasks are introduced. This process facilitates the model's progression from simple to more difficult tasks. Moreover, the proposed method dynamically adjusts the thresholds based on the probability distribution of the highest predictions on unlabeled samples in each class. Thus, the model fully considers the learning difficulty of each class to adapt the thresholds in every training iteration. An overview of the curriculum learning strategy is illustrated in Fig. 4.

B. EA Network

Through the introduction of the AVME-HT feature extraction in Section III, it is evident that the five channels of the proposed features contribute differently to load recognition. Consequently, it is crucial for the model to adaptively assign weight to various channels of the AVME-HT features. Therefore, in this study, the EA network is introduced to enhance neural network representation capabilities for input features by learning channel interrelationships, selectively amplifying or attenuating channels to emphasize vital information. EA network further improves the model's utilization of limited labeled data, enhancing pseudolabeled data quality.

The structure of the proposed model with EA is shown in Fig. 5. Suppose that the input feature ($Z = [z_1, z_2, \ldots, z_c]$) of the EA network is a combination of channels ($z_i \in \mathbb{R}^{w \times 1}$). First, the EA network computed the channel statistics vector via average pooling and maximum pooling layers, aiming to capture the global temporal information of each channel. Subsequently, an encoding–decoding descriptor is employed to learn the statistics vector. The descriptor maps the global features to a lower dimensional potential space and then generates weight vectors. To ensure that the weight vector remains

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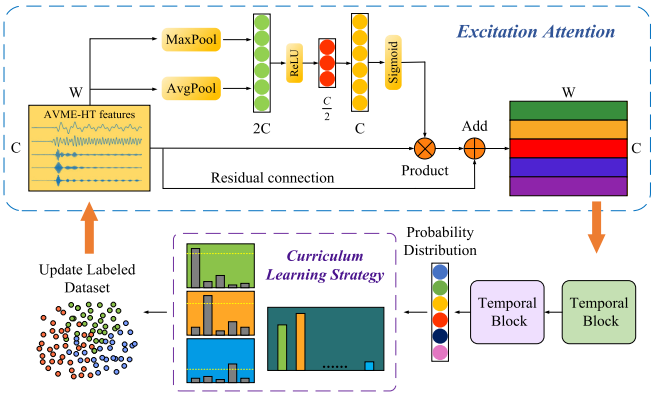


Fig. 5. Proposed model with EA.

within the range of [0, 1], activation functions such as rectified linear unit (ReLU) and sigmoid are applied. The weight vector indicates the importance of each channel. Finally, the weight vector is multiplied elementwise with the input features, allowing for the weighting of different channels within the input feature. This process enhances attention toward significant channels while dampening responses to less significant ones. Additionally, the residual connection was applied to retain the original information, ensuring that the model can still benefit from the full input. This attention mechanism enhances the discriminative power of the model for load identification tasks.

In this study, the classifier was designed using the temporal convolutional network (TCN) structure. The structure comprises two temporal blocks, with each temporal block consisting of two 1-D convolutional layers. A fully connected layer is then connected to the classifier's output, with the size set equal to the number of classes. The selection of TCN was based on its inherent advantages to time series features, including its ability to model long-term dependencies, perform parallel computations, and effectively capture both local and global information.

The loss function is a crucial component of the training process, as it quantifies the discrepancy between the predicted outputs and the true labels. The optimizer is responsible for updating the model's parameters during training, based on the computed gradients. In this study, the loss function was chosen as the cross entropy loss function, while the optimizer was selected as Adam. The max iteration of the SSL network was set to 10. To ensure efficient training and optimal classification, the training epoch was set to 500, and the batch size was set to 32 for each epoch.

V. Experiment

A. Data Preparation

In this study, the plug-load appliance identification dataset (PLAID) was employed, which was obtained at a frequency of 30 kHz. The dataset comprises 1876 records of individually metered appliances from 17 different appliance types [34]. A subset of the PLAID dataset was selected for evaluation, comprising 11 single-metered appliances from various locations, with a total of 1821 measurements. These appliances have various load types, including resistive, inductive, and nonlinear loads.

TABLE I
Loads Selected in Private Dataset

Device	Load type	Device	Load type
Coffee maker	Inductive	Bath heater	Resistive
Electric cooker	Resistive	Washing machine	Inductive
Air fryer	Resistive	Kettle	Resistive
Disinfection cabinet	Resistive	Hair dryer	Resistive
Water heater	Resistive	Electric bicycle	Non-linear
Kitchen hood	Inductive	Induction cooker	Inductive
Blender	Resistive	Fan	Inductive
Purifier	Inductive	Fridge	Inductive
Electric grill	Inductive	LED bulb	Resistive
Clothes steamer	Resistive	Air conditioner	Resistive

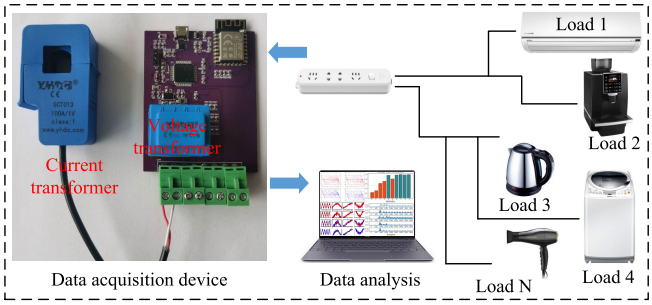


Fig. 6. Diagram of data acquisition device.

Another NILM dataset, the worldwide household and industry transient energy dataset (WHITED) [35], was utilized for evaluating the proposed model. The WHITED dataset recorded the first 5 s of the individually metered appliance start-ups for 110 various appliances in different regions worldwide at a frequency of 44.1 kHz. All 54 kinds of appliances were used in this study as recognition objects, with a total of 1329 measurements.

In addition to the publicly available datasets PLAID and WHITED, a private dataset collected in the laboratory was also utilized. This dataset comprises high-frequency data of 20 distinct types of appliances at a sampling frequency of 10 kHz, acquired through a self-designed data acquisition device, as illustrated in Fig. 6. The appliances included in this dataset are listed in Table I.

B. Experiment Description

In order to evaluate the proposed method, eight comparison models were established. These models include the benchmark model V–I-convolutional neural network (VI-CNN), a model utilizing the AVME-HT feature named AVME-Hilbert transform-cooperative training (AVME-HT-Co-training), as well as four state-of-the-art SSL models: semisupervised smooth teacher graph learning (SSTG) [17], TCN-conditional random field (TCN-CRF) [36], unsupervised data augmentation-average selected with distance (UDA-KD-ADS) [37], and semisupervised multilabel deep learning-TCN (SSML-TCN) [23]. Two models are named AVME-Hilbert transform-CNN (AVME-HT-CNN) and AVME-Hilbert transform-semisupervised curriculum label learning (AVME-HT-SCLL) for ablation experiments. The VI-CNN utilized

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the $V-I$ trajectory image as a feature representation. AVME-HT-Co-training draws inspiration from [19] and consists of two LSTM network structures. In AVME-HT-Co-training, pseudolabeled samples are generated when both networks agree on label predictions. AVME-HT-CNN substitutes the curriculum learning strategy with a supervised CNN, and AVME-HT-SCLL omits the EA network. These eight models were compared against the proposed AVME-HT-EA-SSCL model on the publicly PLAID and WHITED datasets and a private dataset.

In the first experiment, the effectiveness of the proposed AVME method for denoising was evaluated in comparison to the existing signal decomposition methods, EMD, ensemble EMD (EEMD), and VMD. The evaluation involved decomposing the original current signal with a 10 dB SNR into several modes and selecting the top five modes for signal reconstruction based on the energy loss coefficient. The experiment was conducted on 11 types of loads in the PLAID dataset, with 50 groups of current data randomly selected for each load and 10 000 samples intercepted for each data group. In the second experiment, the model's performance was compared with several existing models, VI-CNN, SSTG, TCN-CRF, UDA-KD-ADS, and SSML-TCN, under different noise conditions. Specifically, noise with different SNRs of 20, 30, and 40 dB was first superimposed onto the current signals of three datasets. Subsequently, feature extraction was performed using the current signals with superimposed noise to generate features corresponding to different SNRs. Ultimately, the features were divided into the training set and test set and incorporated into the model for training and inference, respectively. In the third experiment, the classification results of the proposed model and the comparison models were evaluated under different label ratios. In both the second and third experiments, a tenfold cross validation method was used to ensure the stability and reliability of the results. Finally, the ablation experiments were conducted to validate the effectiveness of the proposed method's components, including the attention mechanism and curriculum learning strategy.

To conduct a rigorous comparison of the accuracy of different modal decomposition methods, four metrics were employed to assess the denoising outcomes. The first metric was the mean absolute error (MAE) between the original and denoised data. The second metric was the normalized rms error (NRMS), which gauges the structural similarity between signals. The third metric was the correlation coefficient (CC), defined as

$$\mathrm{CC}(X,Y)=\frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}} \tag{10}$$

where X and Y are the original data and denoising data, respectively. $\mathrm{Cov}(X,Y)$ represents the covariance of X and Y and $\mathrm{var}(\cdot)$ is the variance calculation. The last index is R^2, also known as the coefficient of determination, defined as

$$R^2(X,Y)=1-\frac{\sum(X-Y)^2}{\sum(X-\bar{X})^2} \tag{11}$$

where $\bar{X}$ means the average value of original data X. R_2 reflects the fitting degree of the denoised data relative to the original data. In accordance with [38], the metrics used to assess the performance of load identification methods are accuracy and F1-score. Throughout the article, we use the following abbreviations: induction cooker (IC), hairdryer (HD), compact fluorescent lamp (CFL), air conditioner (AC), bulb (ILB), washing machine (WM), hair iron (HI), water heater (WH), coffee machine (CM), drilling machine (DM), vacuum cleaner (VC), and disinfection cabinet (DC).

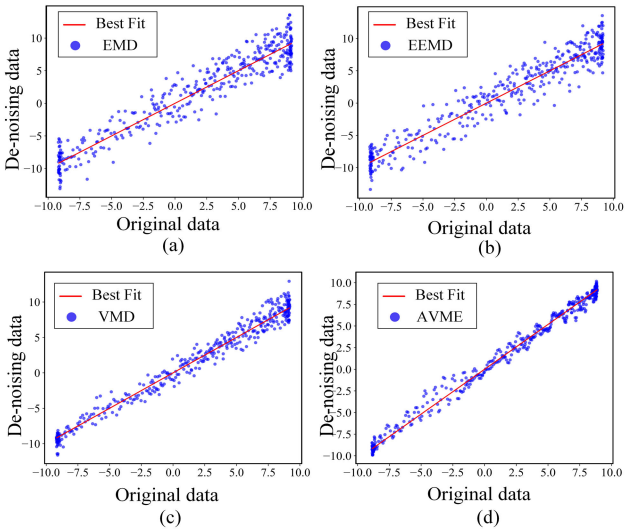


Fig. 7. Original data versus denoising data (scatter plot) at SNR = 10 dB obtained by different methods (a) EMD, (b) EEMD, (c) VMD, and (d) AVME.

VI. Result and Discussion

The experimental results were conducted on a desktop PC equipped with an Intel i7 3.4 GHz CPU and NVIDIA GeForce RTX 3090 GPU. The proposed algorithm was implemented using the PyTorch open-source platform. To leverage the GPU-based parallel computing architecture, the NVIDIA CUDA environment was utilized.

A. Denoising Results of Mode Decomposition Methods

Fig. 7(a)–(d) presents scatter plots comparing denoising performance. The degree of clustering of the points around the best-fit line indicates the effectiveness of the denoising method. In Fig. 7(a), EMD exhibits scattered points, indicating poor denoising similarity. Similarly, Fig. 7(b) suggests that EEMD does not significantly enhance denoising. In contrast, VMD and AVME show tightly clustered points around the best-fit line, indicating superior denoising. Fig. 8 demonstrates AVME's denoising effectiveness compared to EMD, EEMD, and VMD on a 20-dB SNR noisy current signal. The blue, red, and black curves represent noisy, reconstructed, and original signals respectively. AVME closely approximates the original signal, showcasing effective denoising. VMD and VME use the Wiener filter for denoising, adapting better to signal and noise distributions for precise denoising, resulting in better performance.

The four evaluation metrics mentioned above were computed for all the selected data in the PLAID dataset, and the

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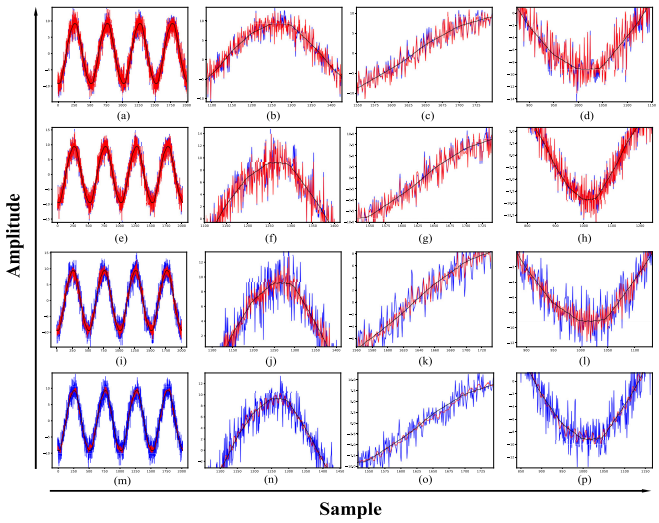


Fig. 8. Blue, red, and black curves represent noise signal at SNR = 10 dB, the reconstructed signal obtained by different methods and the original signal. (a)–(d) EMD, (e)–(h) EEMD, (i)–(l) VMD, and (m)–(p) AVME.

TABLE II

MEAN (AND STANDARD DEVIATION) OF THE FIVE INDICES USED FOR COMPARING DIFFERENT METHODS

	EMD	EEMD	VMD	**AVME**
MAE	1.09 (1.00)	0.96 (0.88)	0.55 (0.51)	0.33 (0.32)
CC	0.94 (0.03)	0.95 (0.02)	0.97 (0.07)	0.99 (0.01)
NRMS	0.35 (0.07)	0.30 (0.07)	0.19 (0.12)	0.11 (0.03)
R^2	0.87 (0.06)	0.90 (0.07)	0.94 (0.11)	0.98 (0.01)
Time(s)	0.49 (0.14)	41.30 (4.41)	3.65 (1.16)	1.25 (0.60)

resulting mean and standard deviation values were presented in Table II. AVME outperformed all other methods in terms of denoising effectiveness, while EMD yielded the poorest results. VMD achieved superior results compared to EMD and EEMD but still fell behind AVME. Table II also provides a comparison of the average time required to process 10 000 samples for each method, revealing that AVME is approximately three times faster than VMD and 33 times faster than EEMD. The prolonged computational time of EEMD is attributed to its requirement to introduce noise to the original signal, generating multiple perturbed signals, and performing EMD on each of these perturbed signals. This process is computationally intensive. The efficiency of AVME lies in directly determining signal harmonic center frequencies using outlier detection, bypassing iterative optimization in the search for these frequencies, thereby reducing computational complexity. Overall, our AVME algorithm offers outstanding denoising performance and computational efficiency, significantly contributing to feature extraction and denoising.

B. Classification Results in Noise and Noise-Free Condition

The efficacy of the proposed AVME-HT-EA-SCLL model in classification tasks under different noise conditions was assessed through experiments conducted in four distinct conditions: a noise-free environment and SNR levels of 20, 30, and 40 dB. First, the noise with different SNRs is superimposed onto the current signals of three datasets. Subsequently, feature extraction is performed using the data with the added noise.

TABLE III

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THREE DATASETS UNDER DIFFERENT NOISE RATIOS

Dataset	Noise (SNR)	**AVME-HT -EA-SCLL**	SSTG [17]	VI-CNN	TCN-CRF [36]	UDA-KD -ADS [37]	SSML-TCN [23]
PLAID dataset	Noise free	96.48%	93.42%	87.20%	96.09%	85.66%	92.46%
	40dB	96.16%	90.31%	80.94%	92.48%	78.59%	88.48%
	30dB	95.32%	83.42%	76.30%	85.09%	71.56%	82.27%
	20dB	94.17%	78.54%	72.25%	77.73%	67.34%	75.96%
WHITED dataset	Noise free	98.49%	98.36%	93.98%	97.84%	90.27%	98.05%
	40dB	98.17%	94.45%	90.92%	93.01%	87.86%	94.27%
	30dB	97.47%	87.82%	83.72%	84.68%	81.53%	86.46%
	20dB	96.24%	82.49%	75.18%	80.05%	73.61%	79.23%
private dataset	Noise free	98.67%	97.25%	92.33%	98.46%	90.64%	96.93%
	40dB	98.33%	92.41%	90.25%	93.73%	86.82%	90.87%
	30dB	98.12%	86.32%	85.92%	87.65%	81.20%	82.59%
	20dB	97.53%	79.47%	78.63%	80.44%	75.31%	77.23%

Finally, the extracted features were divided into the training set and the test set, and incorporated into models for training and inference, respectively. The classification results of the different models are discussed below.

1) Results on PLAID: Table III displays the classification results of different models on the PLAID dataset. The results reveal that the proposed AVME-HT-EA-SCLL model exhibits superior performance over the other models in noise and noise-free conditions. Specifically, in the noise-free condition, the accuracy of AVME-HT-EA-SCLL is improved by 9.28% compared to VI-CNN, reaching 96.48%. The proposed model showed robustness to noise due to the denoising characteristics of AVME-HT features. On the other hand, the SSTG, TCN-CRF, UDA-KD-ADS, and SSML-TCN models demonstrate poor classification results in noise environments. These models achieve significantly decreased accuracies of 14.88%, 18.36%, 19.06%, and 16.50%, respectively, in the SNR of 20 dB compared to the noise-free condition. In particular, the SSTG and TCN-CRF models utilized V–I trajectories and active power, as well as power factors of various harmonics, as input features. These features are derived from the raw voltage and current signals. In cases where the original signals are affected by noise interference, these features may exacerbate the influence of noise, masking the load characteristics and leading to suboptimal classification performance.

Fig. 9(a) illustrates the F1-score results of different models in both noisy and noise-free conditions. The graph clearly demonstrates that our proposed model maintains an excellent F1-score even as the SNR decreases. In contrast, the other models experience a significant decrease in F1-score as the SNR declines. Fig. 10(a) and (b) displays the V–I trajectories on the PLAID dataset in both noise-free and noisy conditions. The blue and red lines represent V–I trajectories generated under these respective conditions. A comparison between Fig. 10(a) and (b) reveals that noise has obscured the fine details of the V–I patterns, leading to reduced discriminative capacity in distinguishing load characteristics.

2) Results on WHITED: The evaluation of the WHITED dataset also yielded similar results. Table III demonstrates that the proposed AVME-HT-EA-SCLL model outperforms the other models in both noisy and noise-free conditions. Notably, the AVME-HT features exhibit remarkable denoising capabilities compared to features such as V–I trajectory and power. Specifically, when the SNR is 20 dB, the accuracy

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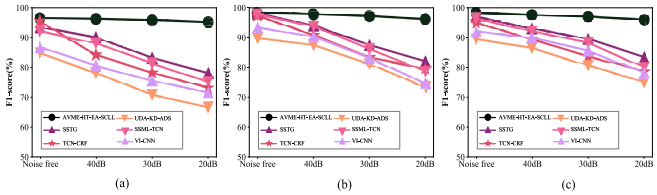


Fig. 9. F1-score result of comparison models in noise and noise-free condition. (a) Results on PLAID dataset. (b) Results on WHITED dataset. (c) Results on private dataset.

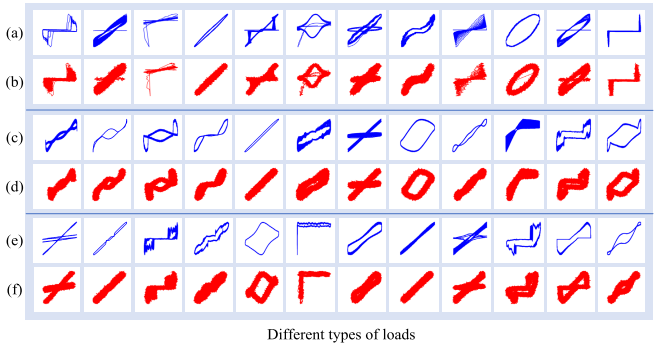


Fig. 10. V–I trajectories of PLAID and WHITED dataset in noise (red) and noise-free (blue) conditions. (a) and (b) V–I trajectories on PLAID dataset. (c) and (d) V–I trajectories on WHITED dataset. (e) and (f) V–I trajectories on private dataset.

of the SSTG, TCN-CRF, UDA-KD-ADS, SSML-TCN, and VI-CNN models was 82.49%, 80.05%, 73.61%, 79.23%, and 75.18%, respectively, while the AVME-HT-EA-SCLL model achieved an accuracy of 96.24%. These results are consistent with the findings in Fig. 9(b), which indicate that the F1-score of the proposed model maintains good performance while the performance of other models degrades as the SNR decreases. Fig. 10(c) and (d) presents the V–I trajectories on the WHITED dataset in the presence and absence of noise. It is evident that noise interference obscures intricate features of the V–I trajectories, resulting in a reduction in the distinguishability of these features.

3) Results on Private Dataset: Table III also presents the classification results of the proposed model and the comparison models in both noisy and noise-free conditions on the private dataset. The proposed model demonstrated superior classification performance, particularly with the AVME-HT feature, which showed less sensitivity to noise. Conversely, the accuracy of the other comparison models decreased as the noise SNR decreased. As depicted in Fig. 10(e) and (f), noise interference blurred the details of the V–I trajectories of different loads in the private dataset, resulting in poorer load identification results.

In summary, our proposed model outperforms other models in various SNR conditions. This superior performance can be attributed to the underlying principle of our AVME-HT feature, which initially denoises the signal using the AVME algorithm before performing feature extraction via the HT. In contrast, models like SSTG and TCN-CRF directly extract features from the noisy original signals, making their extracted features susceptible to noise interference and resulting in poor classification performance.

C. Classification Results Using Different Label Ratios

In this section, the labeled and unlabeled data in each dataset are partitioned into three different proportions. Specifically, labeled data constitutes 30%, 20%, and 10% of the total data, respectively. The proposed model and the comparative model were then evaluated under these three scenarios, respectively.

1) Results on PLAID: The confusion matrix illustrates appliance misclassifications in the comparison experiment on the PLAID dataset, as shown in Fig. 11. It can be observed that compared to other models, the proposed model achieves the highest accuracy, with accuracy rates exceeding 96%. Specifically, VI-CNN and UDA-KD-ADS show accuracy below 90%, leading to confusion between heaters and HDs due to their similar power characteristics. The V–I trajectory feature is an image feature plotted based on the trajectory of load voltage and current variations. Therefore, it struggles to distinguish loads with similar voltage and current characteristics. The UDA-KD-ADS model employs dynamic time warping (DTW) to measure the similarity between time series load signals using Euclidean distances. However, distinguishing between loads of the same type with similar current and power using this approach can be challenging.

Table IV presents the classification results of the proposed AVME-HT-EA-SCLL model and comparison models on the three datasets under different labeling ratios. The results demonstrate the exceptional performance of the proposed model compared to the other models. Notably, the supervised models, AVME-HT-CNN and VI-CNN, exhibit a significant decrease in classification results as the label ratios decreases, with accuracy rates of 43.35% and 31.21% for 182 labeled samples, respectively. In contrast, the proposed AVME-HT-EA-SCLL model maintains a high accuracy of 76.43%. In terms of semisupervised models, the AVME-HT-Co-training model exhibited an 8.16% decline in classification accuracy compared to the proposed model with 182 labeled data. The Co-training model relies on the assumption that the results of two classifiers are largely consistent and provide mutually reinforcing information. However, this assumption might not always hold true in most scenarios, leading to suboptimal results. Fig. 12(d)–(f) displays the F1-score of comparison models on the PLAID dataset with varying labeling ratios. The VI-CNN model achieves an average F1-score below 50% for most loads with only 182 labeled samples. This is due to its reliance on labeled data for training and its incapability to effectively utilize abundant unlabeled data. Consequently, it inadequately learns various load characteristics, leading to suboptimal classification results.

2) Results on WHITED: Table IV presents the classification accuracy of the comparison models across different labeling ratios using the WHITED dataset. The results demonstrate the superior performance of our model. With 132 labels, AVME-HT-CNN and VI-CNN achieved accuracies below 50%. This underscores the challenge faced by supervised models in achieving satisfactory classification performance with limited labeled data. When the proportion of labeled data decreases, semisupervised models such as SSTG and SSML-TCN struggle to achieve accurate classification results. Notably, our

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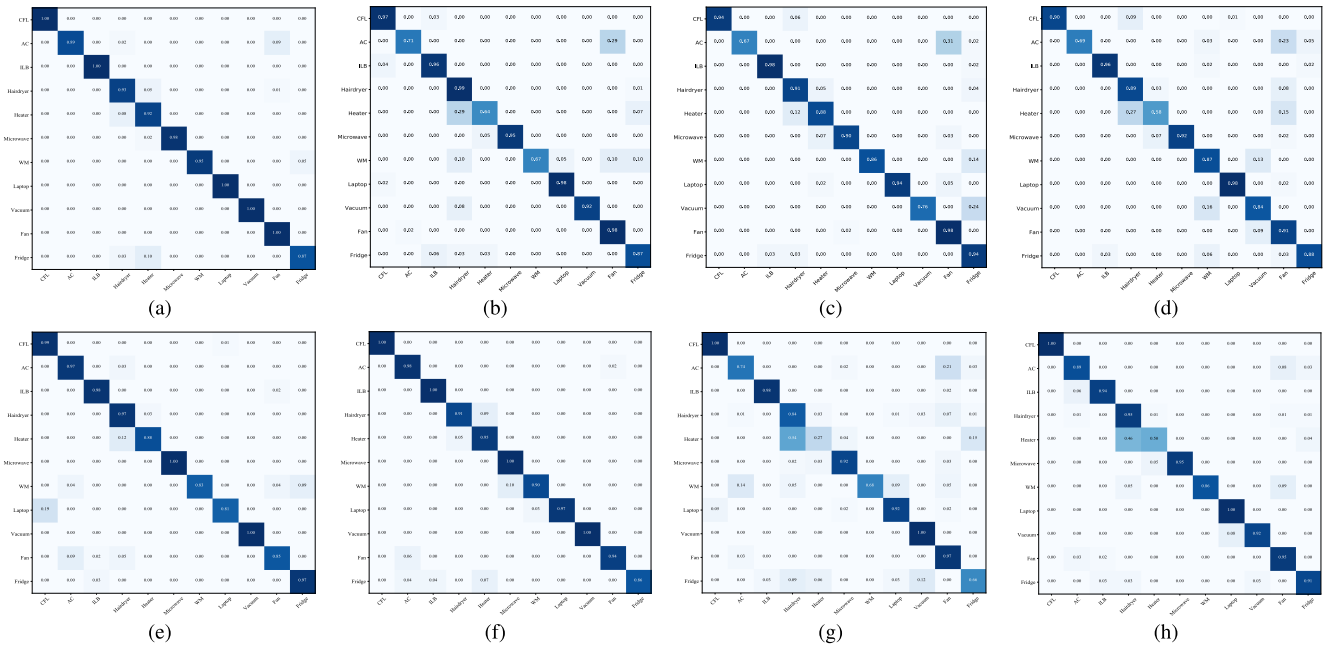


Fig. 11. Confusion matrix of the AVME-HT-EA-SCLL and other comparison models on the PLAID dataset. (a) Result of AVME-HT-EA-SCLL. (b) Result of AVME-HT-Co-training. (c) Result of AVME-HT-CNN. (d) Result of VI-CNN. (e) Result of SSTG. (f) Result of TCN-CRF. (g) Result of UDA-KD-ADS. (h) Result of SSML-TCN.

TABLE IV
CLASSIFICATION ACCURACIES OF EIGHT METHODS ON THREE DATASETS UNDER DIFFERENT LABEL RATIOS

Dataset	Label	**AVME-HT -EA-SCLL**	AVME-HT -Co-training	AVME-HT -CNN	SSTG [17]	VI-CNN	TCN-CRF [36]	UDA-KD -ADS [37]	SSML-TCN [23]
PLAID dataset	All labels	96.48%	91.90%	88.43%	93.42%	87.20%	96.09%	85.66%	92.46%
	546 labels	91.78%	82.65%	73.41%	84.77%	59.53%	90.61%	79.62%	81.57%
	364 labels	78.61%	73.58%	52.02%	79.67%	43.93%	78.44%	75.48%	77.24%
	182 labels	76.43%	68.27%	43.35%	72.59%	31.21%	74.88%	70.13%	71.96%
WHITED dataset	All labels	98.49%	97.74%	95.48%	98.36%	93.98%	97.84%	90.27%	98.05%
	398 labels	90.97%	87.96%	73.68%	88.64%	72.93%	89.27%	82.58%	87.23%
	265 labels	85.71%	75.45%	54.88%	77.96%	50.37%	83.74%	75.46%	76.46%
	132 labels	75.93%	66.17%	42.10%	67.48%	33.83%	73.49%	63.89%	65.23%
Private dataset	All labels	98.67%	96.84%	95.67%	97.25%	92.33%	98.46%	90.64%	96.93%
	300 labels	94.85%	89.86%	83.57%	93.19%	71.74%	92.34%	87.67%	92.52%
	200 labels	91.12%	86.25%	74.50%	89.53%	63.87%	89.20%	85.74%	88.47%
	100 labels	88.78%	83.22%	65.01%	83.84%	52.88%	85.39%	80.93%	81.39%

model achieved an accuracy of 75% with 132 labels, while SSTG and SSML-TCN achieved 67.48% and 65.23% accuracy, respectively. This is attributed to the fact that both the SSTG and SSML-TCN models utilize a teacher–student semisupervised structure, heavily relying on the teacher model, which can propagate errors. The teacher model itself is insufficiently trained on a limited labeled data, leading to inherent inaccuracies. Consequently, the student model replicates these errors, resulting in a decline in the student model's classification performance. Fig. 12(g)–(i) displays the F1-score of the comparison models on the WHITED dataset under different labeling ratios. It can be observed that under 132 labeled data, our model exhibits higher F1-scores on most appliances compared to TCN-CRF and UDA-KD-ADS, such as iron, fan, LED light, and TV. This indicates that our model has an advantage in recognition accuracy over most appliances when compared to other models.

3) Results on Private Dataset: Table IV demonstrates the accuracy of the different models on the private dataset under different labeling ratios. As the labeled data decreases, the classification accuracies of the supervised models AVME-HT-CNN and VI-CNN sharply decline, highlighting the challenge of supervised models with limited labeled data. Furthermore, compared to our model, SSTG experienced a 4.94% decrease in accuracy with 100 labels, indicating that the proposed model based on curriculum learning strategy outperforms the teacher–student model. The SSML-TCN and TCN-CRF achieve accuracies of 81% and 85% respectively with only 100 labeled samples, while the proposed model surpasses them with an accuracy of 88%. Both the proposed model and these two models employ TCN networks, but the proposed model uniquely integrates an EA network that applies attention weights to input feature channels, effectively emphasizing important features and enhancing classification performance. Fig. 12(a)–(c) illustrated the F1-scores of the comparison models on the private dataset across varying labeling ratios. It is evident that with 100 labels, the proposed model exhibited a superior F1-score for most appliances compared to TCN-CRF

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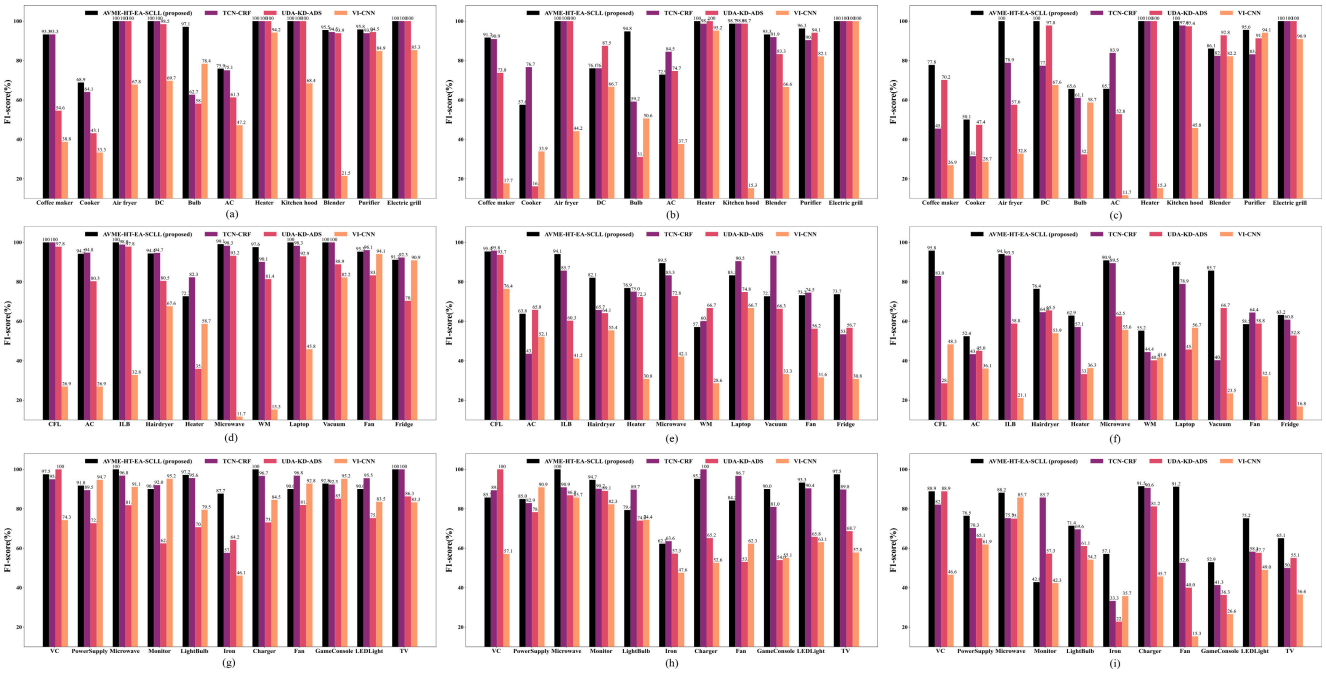


Fig. 12. Comparison-based NILM methods using different label ratios on private, PLAID, and WHITED dataset. (a)–(c) 300, 200, and 100 labels on private dataset. (d)–(f) 546, 364, and 182 labels on PLAID. (g)–(i) 398, 265, and 132 labels on WHITED.

and UDA-KD-ADS, specifically the coffee maker, air fryer, and DC. This demonstrates our model's robust classification performance with limited labeled data.

In conclusion, the classification results indicated that the combination of curriculum learning strategy and EA network effectively improves the recognition accuracy of the proposed model with limited labeled data.

D. Ablation Experiment

The ablation experiments were conducted to validate the effectiveness of our method's components, including the attention mechanism and curriculum learning strategy. Two comparison models called AVME-HT-CNN and AVME-HT-SCLL were designed. AVME-HT-CNN replaces the curriculum learning strategy with a supervised CNN, while AVME-HT-SCLL removes the EA network. Fig. 13 presented radar plots of the classification results across three datasets. It can be observed that, compared to the AVME-HT-SCLL model, our proposed model outperforms in classification under all test conditions. Specifically, on the PLAID dataset, our model achieves classification improvements of 1.76%, 3.91%, and 1.99% under conditions of all labels, 182 labels, and 20-dB SNR noise, respectively. This is attributed to our model's utilization of the EA network to adaptively assign weight to channels within the AVME-HT features, enabling a better understanding of load characteristics within the features and thus enhancing classification results. Furthermore, on the WHITED dataset, our model outperforms AVME-HT-CNN under conditions of 398 labels, 265 labels, and 132 labels, demonstrating improvements of 17.29%, 30.83%, and 33.08%, respectively. This underscores the effectiveness of the curriculum learning strategy in leveraging limited labeled data to enhance classification performance.

E. Discussion

1) Denoising Efficacy and Low Computational Complexity of the AVME-HT Feature: Real-world scenarios in NILM can introduce various sources of noise interference during data collection, such as measurement error or grid disturbance. However, prior studies often assumed ideal, noise-free appliance data for experiments and few studies showcase the denoising capabilities of their methods in the experimental results. Additionally, their methods used in feature extraction, such as VMD, often exhibit high computational complexity.

To address this, our study introduces the AVME-HT feature for simultaneous feature extraction and data denoising. This approach offers two key advantages. First, unlike many feature extraction methods [5], [7], [8], our feature uniquely combines feature extraction and data denoising capabilities. Second, the proposed AVME, employed in our feature, proves more efficient than VMD used in other methods [21], [22]. VMD's high computational complexity arises from its iterative optimization process in determining the central frequency of each decomposed mode. In contrast, AVME efficiently identifies central frequencies using the Pauta criterion and extracts a single frequency mode each time through VME, significantly reducing computational complexity.

2) Role of Curriculum Learning Strategy in SSL: Although many researchers proposed the NILM methods based on SSL, their approaches exhibit notable limitations. SSTG [17] and SSML-TCN [23] employ a teacher–student architecture heavily reliant on the teacher model, potentially leading to error propagation. The inadequately trained teacher model on a limited set of labeled data introduces inherent inaccuracies. In contrast, our study incorporated a curriculum learning strategy into our model, integrating adaptive thresholds during training. Our model dynamically adjusts the thresholds based

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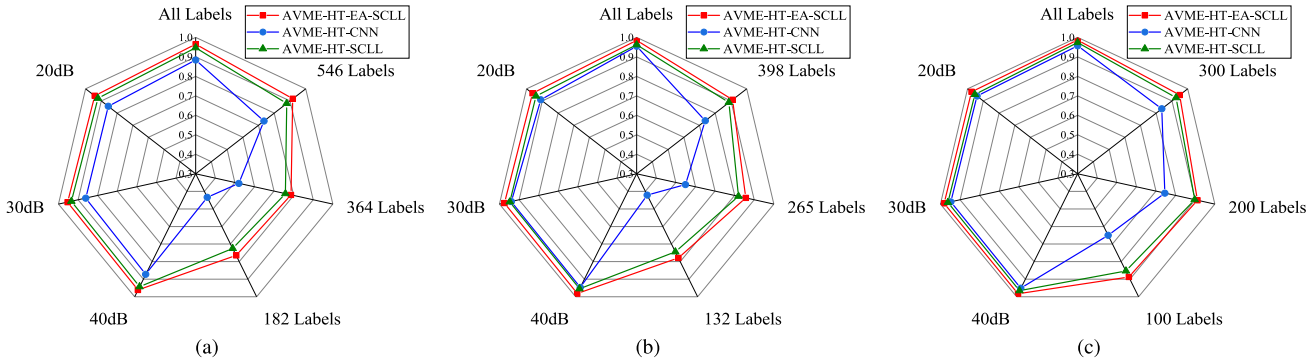


Fig. 13. Radar plots of ablation experiment results. (a) Results on PLAID. (b) Results on WHITED. (c) Results on private dataset.

on the learning difficulty in each class, progressively learning from simpler to more complex tasks. This incremental process enhances the model's understanding of the distributions in the unlabeled data.

The principles of curriculum learning bear similarities to classical mathematical concepts of fuzzy logic, both of which emulate the human brain's thinking process to conduct rule-based reasoning. Some existing methods based on fuzzy rules [39], [40] were utilized for load identification. These methods applied the computed features fuzzified into triangular membership functions to establish fuzzy rules. However, compared with the proposed deep learning model, the intricate nonlinear relationships among load features might be inadequately captured by these fuzzy logic systems. Additionally, the determination of membership function parameters significantly hinges upon domain experts' knowledge, consequently limiting the model's adaptability. Conversely, the curriculum learning strategy dynamically adapts rules based on individual category complexities for pseudolabeling, augmenting the model's adaptability.

3) Limitations and Future Work: In terms of algorithmic computational complexity, this article only explores the comparison within different signal decomposition methods such as the proposed AVME method, EMD, EEMD, and VMD. It emphasizes the low computational complexity of AVME, specifically in the context of feature extraction. However, the overall computational complexity of our model is not thoroughly discussed in the article. Future research will concentrate on lightweight load identification models, refining the AVME-HT-EA-SCLL model to minimize computational complexity and memory requirements while ensuring high classification performance.

VII. Conclusion

The load identification tasks based on deep learning often demand substantial labeled data, which is challenging to meet in practical scenarios. Additionally, the collected data is susceptible to noise interference during the electrical data acquisition. To this end, our work introduces the AVME-HT-EA-SCLL method, a new NILM approach excelling in data denoising and load identification with limited labeled data. Key contributions include three aspects. First, the AVME method efficiently performs signal mode center frequency determination and individual mode extraction, eliminating the need for a complex iterative optimization process. Second, the AVME-HT feature can denoise the signal and extract key information. Third, the AVME-HT-EA-SCLL model utilizes a curriculum learning strategy and an EA network to improve the performance of load identification with limited labeled data. Our model achieves accuracies of 96.48%, 98.49%, and 98.67% on the public datasets PLAID, WHITED, and a private dataset, respectively. Under a 20-dB noise level, our model outperforms comparative models with accuracies of 94.17%, 96.24%, and 97.53%. Even with limited labeled data (182, 132, and 100 samples), our model maintains satisfactory classification accuracy at 76.43%, 75.93%, and 88.78% across all three datasets, showcasing its robust performance.

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